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PRACTICE EXAMPLES

(on where,,having,groupby,orderby)

STEP 1: CREATE TABLE

```
mysql> USE student_db
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> CREATE TABLE orders (
    ->     ord_no INT PRIMARY KEY,
    ->     purch_amt DECIMAL(10,2),
    ->     ord_date DATE,
    ->     customer_id INT,
    ->     salesman_id INT
    -> );
Query OK, 0 rows affected (0.007 sec)
```

STEP 2: INSERT DATA

```
mysql> INSERT INTO orders VALUES
    -> (70001,150.50,'2012-10-05',3005,5002),
    -> (70009,270.65,'2012-09-10',3001,5005),
    -> (70002,65.26,'2012-10-05',3002,5001),
    -> (70004,110.50,'2012-08-17',3009,5003),
    -> (70007,948.50,'2012-09-10',3005,5002),
    -> (70005,2400.60,'2012-07-27',3007,5001),
    -> (70008,5760.00,'2012-09-10',3002,5001),
    -> (70010,1983.43,'2012-10-10',3004,5006),
    -> (70003,2480.40,'2012-10-10',3009,5003),
    -> (70012,250.45,'2012-06-27',3008,5002),
    -> (70011,75.29,'2012-08-17',3003,5007),
    -> (70013,3045.60,'2012-04-25',3002,5001);
Query OK, 12 rows affected (0.009 sec)
Records: 12  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM orders;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.002 sec)
```

QUESTION 1:- WHERE CLAUSE

```
mysql> SELECT *
    -> FROM orders
    -> WHERE purch_amt > 2000;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.001 sec)
```

QUESTION 2:- WHERE CLAUSE

```
mysql> SELECT *
    -> FROM orders
    -> WHERE ord_date = '2012-09-10';
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
+--------+-----------+------------+-------------+-------------+
3 rows in set (0.002 sec)
```

QUESTION 3:- WHERE CLAUSE

```
mysql> SELECT *
    -> FROM orders
    -> WHERE salesman_id = 5001;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
4 rows in set (0.001 sec)
```

QUESTION 4:- ORDER BY

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY purch_amt DESC;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.001 sec)
```

QUESTION 5:- ORDER BY

```
mysql> SELECT *
    -> FROM orders
    -> ORDER BY ord_date;
+--------+-----------+------------+-------------+-------------+
| ord_no | purch_amt | ord_date   | customer_id | salesman_id |
+--------+-----------+------------+-------------+-------------+
|  70013 |   3045.60 | 2012-04-25 |        3002 |        5001 |
|  70012 |    250.45 | 2012-06-27 |        3008 |        5002 |
|  70005 |   2400.60 | 2012-07-27 |        3007 |        5001 |
|  70004 |    110.50 | 2012-08-17 |        3009 |        5003 |
|  70011 |     75.29 | 2012-08-17 |        3003 |        5007 |
|  70007 |    948.50 | 2012-09-10 |        3005 |        5002 |
|  70008 |   5760.00 | 2012-09-10 |        3002 |        5001 |
|  70009 |    270.65 | 2012-09-10 |        3001 |        5005 |
|  70001 |    150.50 | 2012-10-05 |        3005 |        5002 |
|  70002 |     65.26 | 2012-10-05 |        3002 |        5001 |
|  70003 |   2480.40 | 2012-10-10 |        3009 |        5003 |
|  70010 |   1983.43 | 2012-10-10 |        3004 |        5006 |
+--------+-----------+------------+-------------+-------------+
12 rows in set (0.001 sec)
```

QUESTION 6:- SUM()

```
mysql> SELECT SUM(purch_amt) AS total_revenue
    -> FROM orders;
+---------------+
| total_revenue |
+---------------+
|      17541.18 |
+---------------+
1 row in set (0.000 sec)
```

QUESTION 7:- AVG()

```
mysql> SELECT AVG(purch_amt) AS average_order_value
    -> FROM orders;
+---------------------+
| average_order_value |
+---------------------+
|         1461.765000 |
+---------------------+
1 row in set (0.000 sec)
```

QUESTION 8:- MAX()

```
mysql> SELECT MAX(purch_amt) AS highest_order_value
    -> FROM orders;
+---------------------+
| highest_order_value |
+---------------------+
|             5760.00 |
+---------------------+
1 row in set (0.001 sec)
```

QUESTION 9:- MIN()

```
mysql> SELECT MIN(purch_amt) AS lowest_order_value
    -> FROM orders;
+--------------------+
| lowest_order_value |
+--------------------+
|              65.26 |
+--------------------+
1 row in set (0.001 sec)
```

QUESTION 10:- COUNT()

```
mysql> SELECT COUNT(*) AS total_orders
    -> FROM orders;
+--------------+
| total_orders |
+--------------+
|           12 |
+--------------+
1 row in set (0.001 sec)
```

QUESTION 11:- GROUP BY

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5002 |     1349.45 |
|        5001 |    11271.46 |
|        5003 |     2590.90 |
|        5005 |      270.65 |
|        5006 |     1983.43 |
|        5007 |       75.29 |
+-------------+-------------+
6 rows in set (0.001 sec)
```

QUESTION 12:- GROUP BY

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3005 |        1099.00 |
|        3002 |        8870.86 |
|        3009 |        2590.90 |
|        3007 |        2400.60 |
|        3001 |         270.65 |
|        3004 |        1983.43 |
|        3003 |          75.29 |
|        3008 |         250.45 |
+-------------+----------------+
8 rows in set (0.001 sec)
```

QUESTION 13:- GROUP BY + MAX()

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY customer_id;
+-------------+------------------+
| customer_id | highest_purchase |
+-------------+------------------+
|        3005 |           948.50 |
|        3002 |          5760.00 |
|        3009 |          2480.40 |
|        3007 |          2400.60 |
|        3001 |           270.65 |
|        3004 |          1983.43 |
|        3003 |            75.29 |
|        3008 |           250.45 |
+-------------+------------------+
8 rows in set (0.001 sec)
```

QUESTION 14:- GROUP BY + HAVING

```
mysql> SELECT salesman_id,
    ->        SUM(purch_amt) AS total_sales
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING SUM(purch_amt) > 3000;
+-------------+-------------+
| salesman_id | total_sales |
+-------------+-------------+
|        5001 |    11271.46 |
+-------------+-------------+
1 row in set (0.001 sec)
```

QUESTION 15:- GROUP BY + HAVING

```
mysql> SELECT customer_id,
    ->        SUM(purch_amt) AS total_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING SUM(purch_amt) > 2500;
+-------------+----------------+
| customer_id | total_purchase |
+-------------+----------------+
|        3002 |        8870.86 |
|        3009 |        2590.90 |
+-------------+----------------+
2 rows in set (0.001 sec)
```

QUESTION 16:- GROUP BY + HAVING

```
mysql> SELECT customer_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING COUNT(*) > 1;
+-------------+--------------+
| customer_id | total_orders |
+-------------+--------------+
|        3005 |            2 |
|        3002 |            3 |
|        3009 |            2 |
+-------------+--------------+
3 rows in set (0.001 sec)
```

QUESTION 17:- GROUP BY + HAVING + ORDER BY

```
mysql> SELECT customer_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING COUNT(*) > 1
    -> ORDER BY total_orders DESC;
+-------------+--------------+
| customer_id | total_orders |
+-------------+--------------+
|        3002 |            3 |
|        3005 |            2 |
|        3009 |            2 |
+-------------+--------------+
3 rows in set (0.000 sec)
```

QUESTION 18:- BETWEEN + HAVING

```
mysql> SELECT customer_id,
    ->        MAX(purch_amt) AS max_purchase
    -> FROM orders
    -> GROUP BY customer_id
    -> HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;
+-------------+--------------+
| customer_id | max_purchase |
+-------------+--------------+
|        3002 |      5760.00 |
|        3009 |      2480.40 |
|        3007 |      2400.60 |
+-------------+--------------+
3 rows in set (0.001 sec)
```

QUESTION 19:- COUNT + HAVING

```
mysql> SELECT salesman_id,
    ->        COUNT(*) AS total_orders
    -> FROM orders
    -> GROUP BY salesman_id
    -> HAVING COUNT(*) >= 2;
+-------------+--------------+
| salesman_id | total_orders |
+-------------+--------------+
|        5002 |            3 |
|        5001 |            4 |
|        5003 |            2 |
+-------------+--------------+
3 rows in set (0.002 sec)
```

QUESTION 20:- MAX + GROUP BY + HAVING

```
mysql> SELECT ord_date,
    ->        MAX(purch_amt) AS highest_purchase
    -> FROM orders
    -> GROUP BY ord_date
    -> HAVING MAX(purch_amt) > 2000;
+------------+------------------+
| ord_date   | highest_purchase |
+------------+------------------+
| 2012-10-10 |          2480.40 |
| 2012-07-27 |          2400.60 |
| 2012-09-10 |          5760.00 |
| 2012-04-25 |          3045.60 |
+------------+------------------+
4 rows in set (0.001 sec)
```